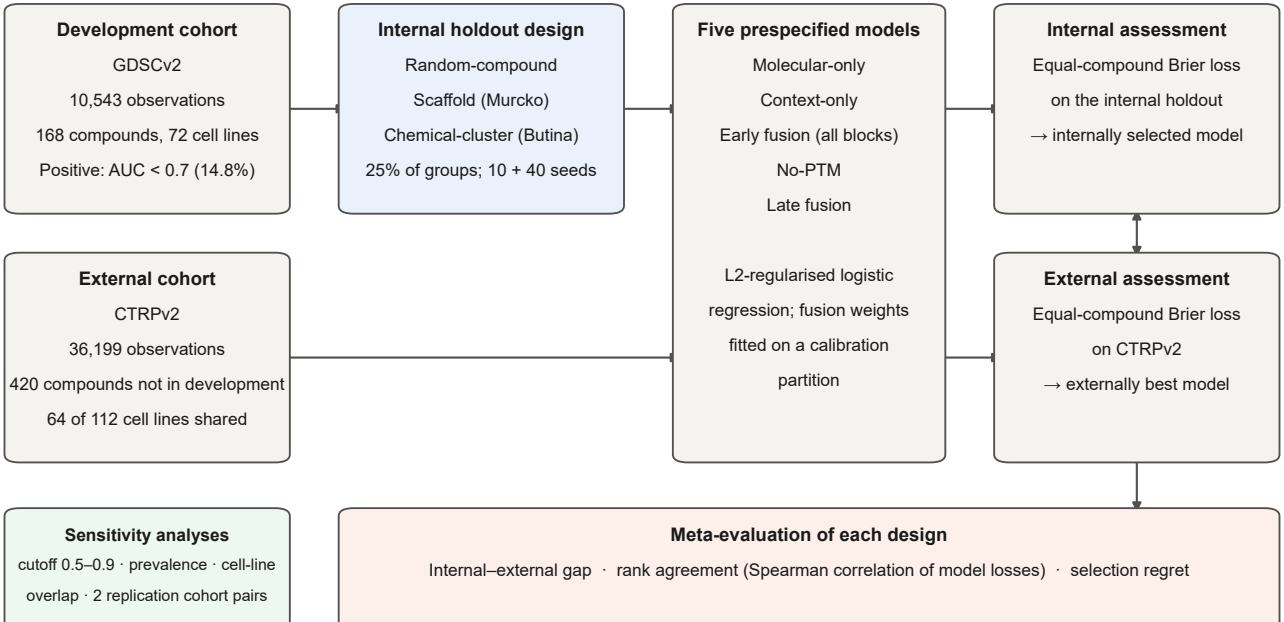
Development cohort
GDSCv2
10,543 observations
168 compounds, 72 cell lines
Positive: AUC < 0.7 (14.8%)
Internal holdout design
Random-compound
Scaffold (Murcko)
Chemical-cluster (Butina)
25% of groups; 10 + 40 seeds
Five prespecified models
Molecular-only
Context-only
Early fusion (all blocks)
No-PTM
Late fusion
L2-regularised logistic regression; fusion weights fitted on a calibration partition
Internal assessment
Equal-compound Brier loss on the internal holdout
→ internally selected model
External cohort
CTRPv2
36,199 observations
420 compounds not in development
64 of 112 cell lines shared
External assessment
Equal-compound Brier loss on CTRPv2
→ externally best model
Sensitivity analyses
cutoff 0.5–0.9 · prevalence · cell-line overlap · 2 replication cohort pairs
Meta-evaluation of each design
Internal–external gap · rank agreement (Spearman correlation of model losses) · selection regret